TORIC 2-FANO MANIFOLDS ARE PROJECTIVE SPACES

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Abstract. We prove the conjecture of Sano, Sato, and Suyama that projective spaces are the only toric 2-Fano manifolds. More generally, a smooth projective toric variety is isomorphic to a projective space if the second Chern character of the tangent bundle has positive degree on all surfaces in the variety.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

For a smooth projective variety X, the second Chern character of T_X is

$$\mathrm{ch}_2(T_X) = \frac{1}{2}\big(c_1(T_X)^2 - 2c_2(T_X)\big).$$

We say that X is 2-*Fano* if $\mathrm{ch}_2(T_X)$ is ample, i.e. $\mathrm{ch}_2(T_X)\cdot\alpha > 0$ for any non-zero class α in the closure of the real convex cone generated by the classes of effective 2-cycles on X [AC13, §1, p. 2]. When X is toric, the cone generated by the classes of surfaces in X is polyhedral [Sat12, Proposition 2.4], so 2-Fano is equivalent to ch_2-positive (cf. [dJS06, Definition 1.2]), that is, $\mathrm{ch}_2(T_X)\cdot S > 0$ for any closed surface $S \subset X$.

Remark 1.1. We remark that our definition of 2-Fano has a slight difference with the original definition [dJS07, Definition 1.1], where $\mathrm{ch}_2(T_X)$ is only assumed to be nef. Note that $(\mathbb{P}^1)^d$ is 2-Fano in the sense of [dJS07, Definition 1.1] but not 2-Fano in the sense of [AC13, §1, p. 2]. We will not use [dJS07, Definition 1.1] in the rest of this paper.

Several classes of 2-Fano manifolds, not necessarily toric, are classified. The only 2-Fano threefolds are $\mathbb{P}^3$ and the smooth quadric hypersurface in $\mathbb{P}^4$ [AC13, Theorem 2]. Moreover, 2-Fano manifolds X with $\dim X \geq 3$ and $-K_X \sim iH$ for an ample Cartier divisor H and an integer $i \geq \dim X - 2$ are classified in [AC13, Theorem 3]. The 2-Fano rational homogeneous spaces of Picard number one are classified in [Ara+22, Theorem 1.3], and the non-homogeneous 2-Fano horospherical varieties of Picard number one in [AC24, Theorem 1.1]. All known 2-Fano manifolds X have Picard number $\rho(X) = 1$ [Ara+23, §1], and [Cas24, p. 850, Question] asks whether all 2-Fano manifolds have Picard number one.

We now turn to toric varieties. Among the known examples, the only toric 2-Fano manifolds are projective spaces [Nob11, §1]. Sano, Sato, and Suyama proposed the following conjecture.

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Conjecture 1.2 ([SSS21, Conjecture 4.3]). *Let X be a smooth projective toric Fano variety of dimension $d \geq 1$. Assume that X is 2-Fano. Then $X \cong \mathbb{P}^d$.*

See also [Ara+23, Conjecture 1.1] and [Ara+26, Conjecture 1.1]. Conjecture 1.2 was proven in the following cases:

(1) When $d \leq 8$ [SSS21, Theorem 1.2]. For $d = 2$ see [AC13, §2, p. 5], for $d = 3$ see [AC13, Theorem 2], and for $d = 4$ see [Nob11, Theorem 4.1].
(2) When $\rho(X) \leq 3$, even without assuming that $-K_X$ is ample [SS20, Corollary 1.2].
(3) When X has an extremal contraction of fiber type, even without assuming that $-K_X$ is ample [Sat16, Theorem 3.1(a)].
(4) When d is even and X is the smooth toric Fano d-fold whose fan has primitive ray generators $\pm e_1, \ldots, \pm e_d, \pm(e_1 + \cdots + e_d)$, or the smooth toric Fano d-fold whose fan has primitive ray generators $\pm e_1, \ldots, \pm e_d, -(e_1 + \cdots + e_d)$, where $e_1, \ldots, e_d$ is a basis of the lattice of one-parameter subgroups of the torus of X. In both cases X is not ch_2-positive [SSS21, Theorem 3.1].

[SSS21, Theorem 1.2] is proved by a computer check over the list of smooth toric Fano d-folds, one dimension at a time. For $4 \leq d \leq 8$ and any smooth toric Fano d-fold X with $\rho(X) \geq 2$, the check finds a torus-invariant surface $S \subset X$ with $\rho(S) = 2$ and $\mathrm{ch}_2(T_X) \cdot S \leq 0$, except when d is even and X is the variety in (4) whose fan has primitive ray generators $\pm e_1, \ldots, \pm e_d, \pm(e_1 + \cdots + e_d)$ [SSS21, Propositions 4.1, 4.2]. Such a dimension-by-dimension check does not extend to all dimensions.

Further cases of Conjecture 1.2 were proven by means of primitive collections. Let X be a smooth projective toric variety of dimension d. A primitive collection of the fan of X is a set P of primitive ray generators of the fan such that P does not generate a cone of the fan, while $P \backslash \{x\}$ does for any $x \in P$ [Bat91, Definition 2.6]. A primitive collection P is called *centered* if $\sum_{x \in P} x = 0$; these are the centrally symmetric primitive collections of [Ara+23, Definition 2.3]. The fan of X has a centered primitive collection [Bat91, Proposition 3.2] (see [CFH14, Corollary 3.3] for a geometric proof). Let $m(X)$ be the smallest positive integer m such that the fan of X has a centered primitive collection with $m+1$ elements [Ara+23, Definition 2.10]. Since primitive ray generators are non-zero, and the elements of a proper subset of a primitive collection are linearly independent, $1 \leq m(X) \leq d$. Conjecture 1.2 holds if $m(X) = 1$ [Ara+23, Theorem 1.2], if $m(X) \geq d - 2$ [Ara+23, Corollary 1.5], and if $d > 2$ and $m(X) = 2$ [Ara+26, Theorem 1.2].

The results for $m(X) = 1$ and $m(X) = 2$ follow the strategy proposed in [Ara+23, §1]. Let P be a centered primitive collection with $m + 1$ elements. Then the dense torus-invariant open subset $U \subset X$ of [Ara+23, Proposition 2.13] admits a $\mathbb{P}^m$-bundle structure over a smooth toric variety W. If W contains a rational curve, then U contains a surface S with $\mathrm{ch}_2(T_X) \cdot S \leq 0$ [Ara+26, Proposition 5.2], and such a curve exists if $m < d$ and $\mathrm{codim}_X(X \backslash U) \geq 2$ [Ara+26, Lemma 5.1]. If $X \backslash U$ contains a prime divisor, the strategy has three steps [Ara+26, §1]. First, toric blowdowns and flips turn X into a complete smooth toric variety Y with a $\mathbb{P}^m$-bundle structure on an open subset whose complement has codimension at least two. Second, by [Ara+26, Lemma 5.1 and Proposition 5.2] applied to Y, there exists a surface $S_Y \subset Y$ with $\mathrm{ch}_2(T_Y) \cdot S_Y \leq 0$. Third, one compares $\mathrm{ch}_2(T_Y) \cdot S_Y$ with $\mathrm{ch}_2(T_X) \cdot \tilde{S}$, where $\tilde{S} \subset X$ is the strict transform of S_Y, to show that $\mathrm{ch}_2(T_X) \cdot \tilde{S} \leq 0$.

The strategy takes P with $m = m(X)$. For $m(X) = 2$, the first step already needs flips, Y may be non-projective [Ara+26, §1], and [Ara+26, Corollary 6.7] compares ch_2 along the sequence of blowdowns and flips. For $m(X) = 3$, the blowdowns and flips in the first step can lead to singular varieties, already for some smooth toric Fano 6-folds of Picard number four [Ara+26, Examples B.1, B.2]. According to the abstract of [Ara+26], going beyond $m(X) = 2$ needs either a much finer combinatorial analysis of primitive collections or a fundamentally new approach.

By [Ara+26, Corollary 1.3], a counterexample X to Conjecture 1.2 would satisfy

$$d \geq 9, \quad 3 \leq m(X) \leq d-3, \quad \text{and} \quad 4 \leq \rho(X) < 2d - \frac{\sqrt{60d+1249}-37}{30}.$$

The goal of this paper is to prove Conjecture 1.2.

Theorem 1.3. *Let X be a smooth projective toric variety of dimension $d \geq 1$. Assume that $\mathrm{ch}_2(T_X) \cdot S > 0$ for any surface $S \subset X$. Then $X \cong \mathbb{P}^d$.*

Since toric 2-Fano manifolds satisfy the assumptions of Theorem 1.3, Conjecture 1.2 holds, and projective spaces are the only toric 2-Fano manifolds. In particular, toric 2-Fano manifolds have Picard number one, which answers [Cas24, p. 850, Question] affirmatively in the toric case.

Sketch of the proof. By Lemma 2.4, the fan of X has a centered primitive collection $P = \{x_0, \dots, x_m\}$ with $1 \leq m \leq d$, and $X \cong \mathbb{P}^d$ if $m = d$. If $m \leq d-1$, then X contains a surface S with $\mathrm{ch}_2(T_X) \cdot S \leq 0$ (Proposition 5.1), which contradicts the assumption of Theorem 1.3. The proof of Proposition 5.1 has three steps.

(1) Construction-Proposition 4.4 resolves the normalization of the closure in X of an $(m+1)$-dimensional subtorus containing the one-parameter subgroups given by P. This yields a smooth projective toric variety X' and a toric morphism $f : X' \to X$ with $f^*D_x = D'_x$ for any $x \in P$, where $D_x \subset X$ and $D'_x \subset X'$ are the torus-invariant prime divisors corresponding to x.

(2) For the torus-invariant surfaces $S'_{ij} = V(\mathrm{Cone}(P \backslash \{x_i, x_j\})) \subset X'$ with $0 \leq i < j \leq m$, [CLS11, Proposition 13.1.2(a)] and the projection formula for the birational morphisms $S'_{ij} \to f(S'_{ij})$ (Construction-Proposition 4.4(4)) imply

$$2\,\mathrm{ch}_2(T_X) \cdot f(S'_{ij}) = \sum_{x \in P} \left((D'_x)^2 \cdot S'_{ij}\right) + \sum_{v \notin P} \left((f^*D_v)^2 \cdot S'_{ij}\right),$$

where v runs over the primitive ray generators of the fan of X. For any $v \notin P$, $((f^*D_v)^2 \cdot S'_{ij}) \leq 0$ by Lemma 2.5, since $(f^*D_v \cdot F'_{ij}) = 0$ for a general fiber F'_{ij} of a toric morphism $S'_{ij} \to \mathbb{P}^1$ (Construction-Propositions 3.2(4)(5) and 4.4(3)).

(3) On a single S'_{ij}, the sign of $\sum_{x \in P}((D'_x)^2 \cdot S'_{ij})$ is not determined, but after summing over all $m(m+1)/2$ pairs, Proposition 3.3 implies

$$\sum_{0 \leq i < j \leq m} \sum_{x \in P} \left((D'_x)^2 \cdot S'_{ij}\right) \leq 0.$$

Thus $\sum_{0 \leq i < j \leq m} \mathrm{ch}_2(T_X) \cdot f(S'_{ij}) \leq 0$, and some $f(S'_{ij})$ is the desired surface. Lemma 2.4 depends on [Bat91, Proposition 3.2], and step (2) combines the projection formula with Lemma 2.5, a consequence of the Hodge index theorem, while steps (1) and (3) are new.

Steps (1) and (3) depend on the following two ideas.

(a) Closures of subtori. When the complement of the open subset of X with the $\mathbb{P}^m$-bundle structure given by P [Ara+23, Proposition 2.13] contains a prime divisor, the strategy of [Ara+23, §1] first modifies X by toric blowdowns and flips [Ara+26, §1], which can lead to singular varieties for $m(X) = 3$ [Ara+26, Examples B.1, B.2]. Instead of modifying X, Construction-Proposition 4.1 uses the normalization of the closure of the subtorus in step (1). Any primitive ray generator of the fan of the normalization not in P is contained in the relative interior of a cone of the fan of X without rays in P, so the pullback of D_x to the normalization is the prime divisor corresponding to x for any $x \in P$. The normalization can be singular (Example 4.2), and Lemma 4.3 resolves it by star subdivisions only at vectors in cones without rays in P, which keeps the pullbacks of the D_x prime.

(b) Summing over all pairs. In the proof of [Ara+26, Proposition 5.2], the surfaces are $\mathbb{P}^1$-bundles over $\mathbb{P}^1$, and the pair i, j is taken so that the corresponding summands in a splitting of a vector bundle whose projectivization is the $\mathbb{P}^m$-bundle over a rational curve have the smallest degrees.

The surfaces S'_{ij} carry no such data, and Proposition 3.3 sums over all pairs instead. For the curves $C'_i = V(\mathrm{Cone}(P\backslash\{x_i\})) \subset X'$, the sum $\sum_{x\in P}((D'_x)^2 \cdot S'_{ij})$ splits into the terms

$$((D'_x|_{S'_{ij}} - \frac{1}{2}(C'_i + C'_j))^2)_{S'_{ij}} \leq 0$$

(Construction-Proposition 3.2(2)(5) and Lemma 2.5) and a linear combination of the intersection numbers $(D'_x \cdot C'_i)$ and $(D'_x \cdot C'_j)$ on X', whose sign is not determined. Since $\frac{1}{2}(C'_i + C'_j)$ is symmetric in i and j, and

$$\sum_{0\leq i<j\leq m} \left((C'_i - C'_j)^2\right)_{S'_{ij}} = \sum_{i=0}^{m}\sum_{x\in P}(D'_x \cdot C'_i)$$

by Construction-Proposition 3.2(1)–(3), the linear combinations add up to

$$\frac{3m-1}{4}\sum_{0\leq i<j\leq m}((C'_i - C'_j)^2)_{S'_{ij}}$$

after summation over all pairs, and $((C'_i - C'_j)^2)_{S'_{ij}} \leq 0$ by Lemma 2.5.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce centered primitive collections. In Section 3, for a smooth projective toric variety X whose fan has a centered primitive collection with exactly $\dim X$ elements, we study the torus-invariant surfaces in X corresponding to the cones generated by all but two elements of the collection, and prove Proposition 3.3. In Section 4, for a smooth projective toric variety X whose fan has a centered primitive collection with at most $\dim X$ elements, we use the closure in X of a subtorus of the torus of X to construct a smooth projective toric variety whose dimension equals the number of elements of the collection, together with a toric morphism to X (Construction-Proposition 4.4). In Section 5 we prove Proposition 5.1 and Theorem 1.3.

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2. Preliminaries

2.1. Toric varieties.

Definition 2.1. Let N be a lattice, $T_N := N \otimes_{\mathbb{Z}} \mathbb{C}^*$ the torus whose lattice of one-parameter subgroups is N, $M := \mathrm{Hom}(N, \mathbb{Z})$ the dual lattice of N, and $N_{\mathbb{R}} := N \otimes_{\mathbb{Z}} \mathbb{R}$. For a subset $A \subset N_{\mathbb{R}}$, we denote by $\mathrm{Cone}(A)$ and $\mathrm{Span}(A)$ the convex cone and the real linear subspace generated by A, respectively. We also write $\mathrm{Cone}(v_1, \ldots, v_k)$ for $\mathrm{Cone}(\{v_1, \ldots, v_k\})$ when $v_1, \ldots, v_k \in N_{\mathbb{R}}$.

Definition 2.2. For a fan Σ in $N_{\mathbb{R}}$, we denote by X_Σ the toric variety associated to Σ [CLS11, Theorem 3.1.5], and by $G(\Sigma)$ the set of primitive generators of the rays of Σ. For a cone $\sigma \in \Sigma$, we denote by $G(\sigma)$ the set of primitive generators of the rays of σ. For a cone $\tau \in \Sigma$, we denote by $V(\tau)^\circ$ the T_N-orbit in X_Σ corresponding to τ [CLS11, Theorem 3.2.6(a)], and by $V(\tau)$ the closure of $V(\tau)^\circ$ in X_Σ. For $v \in G(\Sigma)$, we denote by $D_v := V(\mathrm{Cone}(v))$ the torus-invariant prime divisor on X_Σ corresponding to v.

2.2. Primitive collections.

We recall primitive collections of smooth fans [Bat91, Definition 2.6] and centered primitive collections, which are called centrally symmetric in [Ara+23, Definition 2.3].

Definition 2.3. Let Σ be a smooth fan in $N_{\mathbb{R}}$. A subset $P \subset G(\Sigma)$ is called a *primitive collection* of Σ if $\mathrm{Cone}(P) \notin \Sigma$ and $\mathrm{Cone}(P\backslash\{x\}) \in \Sigma$ for any $x \in P$. We say that a primitive collection P of Σ is *centered* if $\sum_{x\in P} x = 0$.

In Section 3, we use the condition that P is centered to construct toric morphisms from certain torus-invariant surfaces of X_Σ to $\mathbb{P}^1$.

Lemma 2.4. *Let X be a smooth projective toric variety of dimension $d \geq 1$ with fan Σ in $N_\mathbb{R}$. Then the following hold.*

(1) Σ has a centered primitive collection.
(2) For any centered primitive collection P of Σ with $m+1$ elements, we have $1 \leq m \leq d$, and $P\backslash\{x\}$ is part of a basis of N for any $x \in P$.
(3) If Σ has a centered primitive collection with $d+1$ elements, then $X \cong \mathbb{P}^d$.

Proof. (1) By [Bat91, Proposition 3.2] (see also [CFH14, Corollary 3.3]), there exists a primitive collection P of Σ in the sense of [Bat91, Definition 2.6] such that $\sum_{x\in P} x = 0$. By [Bat91, Definition 2.6], $\mathrm{Cone}(P\backslash\{x\}) \in \Sigma$ for any $x \in P$. For any $x \in P$, we have $-x = \sum_{y\in P\backslash\{x\}} y$, which belongs to $\mathrm{Cone}(P)$, so $\mathrm{Cone}(P)$ contains the line $\mathbb{R}x$ and is not strongly convex. Thus $\mathrm{Cone}(P) \notin \Sigma$, and P is a centered primitive collection of Σ.

(2) Let P be a centered primitive collection of Σ with $m+1$ elements. Since the elements of P are non-zero and $\sum_{y\in P} y = 0$, the set P has at least two elements, that is, $m \geq 1$. Let $x \in P$. By Definition 2.3, $\sigma := \mathrm{Cone}(P\backslash\{x\}) \in \Sigma$, which is a smooth cone, so $P\backslash\{x\} \subset G(\sigma)$ is part of a basis of N [CLS11, Theorem 3.1.19(a)]. In particular, $m \leq \operatorname{rank} N = \dim X = d$.

(3) Let $P = \{x_0, \ldots, x_d\}$ be a centered primitive collection of Σ with $d+1$ elements. By (2), for any $x \in P$, the set $P\backslash\{x\}$ is part of a basis of N, hence a basis of N, since it has $d = \operatorname{rank} N$ elements. Since P is centered, $x_0 = -\sum_{k=1}^d x_k$. Let $\Sigma_P := \{\mathrm{Cone}(Q) \mid Q \subsetneq P\}$. By [CLS11, Example 3.1.10], Σ_P is a fan in $N_\mathbb{R}$ and $X_{\Sigma_P} \cong \mathbb{P}^d$, so Σ_P is complete by [CLS11, Theorem 3.1.19(c)].

For any $Q \subsetneq P$ and $x \in P\backslash Q$, the cone $\mathrm{Cone}(Q)$ is a face of $\mathrm{Cone}(P\backslash\{x\}) \in \Sigma$, so $\Sigma_P \subset \Sigma$. Since Σ_P is complete, $\Sigma = \Sigma_P$, and $X \cong X_{\Sigma_P} \cong \mathbb{P}^d$. □

2.3. **A consequence of the Hodge index theorem.** The toric morphisms from torus-invariant surfaces to $\mathbb{P}^1$ in Sections 3 and 5 can have reducible fibers, which happens when the fan of the surface has more than four rays. Concerning such fibers, we only use the following lemma.

Lemma 2.5. *Let X be a smooth projective surface, $\pi : X \to \mathbb{P}^1$ a surjective morphism, F a general fiber of π, and D a $\mathbb{Q}$-divisor on X. Assume that $(D \cdot F) = 0$. Then $(D^2) \leq 0$.*

Proof. Since fibers of π over distinct points of $\mathbb{P}^1$ are linearly equivalent and disjoint, we have $(F^2) = 0$. Let A be an ample divisor on X. Then $(F \cdot A) > 0$. Since $(cD \cdot F) = c(D \cdot F)$ and $((cD)^2) = c^2(D^2)$ for any integer $c > 0$, possibly replacing D with a positive multiple, we may assume that D is a divisor. Let $D_1 := (D \cdot A)F - (F \cdot A)D$. Then $(D_1 \cdot A) = 0$, so $(D_1^2) \leq 0$ by the Hodge index theorem [Har77, Ch. V, Theorem 1.9]. Since $(F^2) = 0$ and $(D \cdot F) = 0$, we have

$$(D_1^2) = (D \cdot A)^2(F^2) - 2(D \cdot A)(F \cdot A)(D \cdot F) + (F \cdot A)^2(D^2) = (F \cdot A)^2(D^2).$$

Thus $(D^2) \leq 0$, as $(F \cdot A) > 0$. □

3. Toric surfaces over the projective line

In this section, we prove Proposition 3.3, which is applied in Section 5 to the variety constructed in Section 4.

Set-up 3.1. Let X, Σ, and P be as follows.

(1) X is a smooth projective toric variety of dimension $m+1$ with fan Σ in $N_\mathbb{R}$, where $m \geq 1$ is an integer.
(2) $P = \{x_0, \ldots, x_m\}$ is a centered primitive collection of Σ.

For any $0 \leq i < j \leq m$, let $\tau_{ij} := \mathrm{Cone}(P\backslash\{x_i, x_j\})$. Let $S_{ij} := V(\tau_{ij})$. For any $0 \leq i \leq m$, let $C_i := V(\mathrm{Cone}(P\backslash\{x_i\}))$.

The assumption on the dimension of X in Set-up 3.1 ensures that each S_{ij} is a surface (Construction-Proposition 3.2(1)).

In general, S_{ij} is not a $\mathbb{P}^1$-bundle over $\mathbb{P}^1$, since the fan $\mathrm{Star}(\tau_{ij})$ of S_{ij} [CLS11, Proposition 3.2.7] may have more than four rays. Instead, Proposition 3.3 uses a surjective toric morphism from S_{ij} to $\mathbb{P}^1$ and the intersection numbers of the prime divisors D_v with a general fiber of the morphism, which Construction-Proposition 3.2 provides. Since P is centered, the images of x_i and x_j in $N/(N \cap \mathrm{Span}(\tau_{ij}))$ are opposite, and the morphism to $\mathbb{P}^1$ is induced by the projection along the line spanned by them: the kernel of q_{ij} in Construction-Proposition 3.2 contains the image of x_i, a condition that determines q_{ij} up to sign.

Construction-Proposition 3.2. Notation and conditions as in Set-up 3.1. Let i, j be integers such that $0 \le i < j \le m$, and let $N_{ij} := N/(N \cap \mathrm{Span}(\tau_{ij}))$. Then there exists a surjective homomorphism $q_{ij} : N_{ij} \to \mathbb{Z}$ whose kernel contains the image of x_i, and for any such q_{ij} the following hold.

(1) S_{ij} is a smooth projective toric surface containing the curves C_i and C_j, and $C_i \cap C_j = \emptyset$.
(2) $D_{x_j}|_{S_{ij}} = C_i$ and $D_{x_i}|_{S_{ij}} = C_j$.
(3) $(D_{x_i} \cdot C_i) = (D_{x_j} \cdot C_j) = 0$.
(4) q_{ij} induces a surjective toric morphism $\pi_{ij} : S_{ij} \to \mathbb{P}^1$.
(5) For a general fiber F_{ij} of π_{ij} and any $v \in G(\Sigma)$, we have $(D_v \cdot F_{ij}) = 1$ if $v \in P$, and $(D_v \cdot F_{ij}) = 0$ if $v \notin P$.

Proof. For $w \in N$, let $\bar{w}$ be the image of w in N_{ij}. We first show that q_{ij} exists. By Lemma 2.4(2), the set $P\backslash\{x_j\} = (P\backslash\{x_i, x_j\}) \cup \{x_i\}$ is part of a basis of N. Thus $G(\tau_{ij}) = P\backslash\{x_i, x_j\}$, $N \cap \mathrm{Span}(\tau_{ij})$ is generated by $G(\tau_{ij})$, and N_{ij} is a lattice of rank 2 such that $\bar{x}_i$ is part of a basis of N_{ij}. Hence $N_{ij}/\mathbb{Z}\bar{x}_i \cong \mathbb{Z}$, and the quotient map $N_{ij} \to N_{ij}/\mathbb{Z}\bar{x}_i \cong \mathbb{Z}$ is a surjective homomorphism whose kernel contains $\bar{x}_i$.

Fix a surjective homomorphism $q_{ij} : N_{ij} \to \mathbb{Z}$ whose kernel contains $\bar{x}_i$. Since q_{ij} is surjective and N_{ij} is of rank 2, the kernel of $q_{ij} \otimes \mathbb{R}$ is the line $\mathbb{R}\bar{x}_i$. Since $\sum_{k=0}^{m} x_k = 0$ and $x_k \in \tau_{ij}$ for any $k \neq i, j$, we have $\bar{x}_j = -\bar{x}_i$.

(1) By Definition 2.3 and Lemma 2.4(2), τ_{ij} is a common face of the m-dimensional smooth cones $\mathrm{Cone}(P\backslash\{x_i\})$ and $\mathrm{Cone}(P\backslash\{x_j\})$ of Σ, so S_{ij} is a surface containing the curves C_i and C_j [CLS11, Theorem 3.2.6]. By [CLS11, Proposition 3.2.7], S_{ij} is isomorphic to the toric variety of the fan $\mathrm{Star}(\tau_{ij})$ in $N_{ij} \otimes_{\mathbb{Z}} \mathbb{R}$. Moreover, S_{ij} is projective since it is closed in X. For any $\sigma \in \Sigma$ having τ_{ij} as a face, $G(\sigma)$ is part of a basis of N [CLS11, Theorem 3.1.19(a)] and contains $G(\tau_{ij})$. Since $N \cap \mathrm{Span}(\tau_{ij})$ is generated by $G(\tau_{ij})$, the images of the elements of $G(\sigma)\backslash G(\tau_{ij})$ in N_{ij} form part of a basis of N_{ij}. Thus any cone of $\mathrm{Star}(\tau_{ij})$ is smooth, and S_{ij} is smooth by [CLS11, Theorem 3.1.19(a)]. Finally, no cone $\sigma \in \Sigma$ contains P: otherwise $-x_0 = x_1 + \cdots + x_m \in \sigma$, so σ contains the line $\mathbb{R}x_0$, which contradicts the strong convexity of σ. Since $(P\backslash\{x_i\}) \cup (P\backslash\{x_j\}) = P$, no cone of Σ contains both $\mathrm{Cone}(P\backslash\{x_i\})$ and $\mathrm{Cone}(P\backslash\{x_j\})$. Thus $C_i \cap C_j = \emptyset$ by [CLS11, Theorem 3.2.6(d)].

(2) Let $v \in G(\Sigma)\backslash G(\tau_{ij})$. We show that

$$D_v|_{S_{ij}} = V(\tau_{ij} + \mathrm{Cone}(v)) \ \text{ if } \ \tau_{ij} + \mathrm{Cone}(v) \in \Sigma, \quad \text{and} \quad D_v \cap S_{ij} = \emptyset \ \text{ otherwise.} \tag{3.1}$$

Let $\sigma \in \Sigma$ be a cone having τ_{ij} as a face, let $r := \dim \sigma$, and let $U_\sigma \subset X$ be the affine open subset corresponding to σ [CLS11, Theorem 3.2.6(c)]. We extend $G(\sigma)$ to a basis of N. By [CLS11, Example 1.2.21], the characters χ^u, for u in the dual basis of M, are coordinates on $U_\sigma \cong \mathbb{C}^r \times (\mathbb{C}^*)^{m+1-r}$. By [CLS11, Proposition 4.1.2] and [CLS11, Theorem 3.2.6(c)(d)], for any $w \in G(\Sigma)$, $D_w \cap U_\sigma$ is the divisor of the coordinate corresponding to w if $w \in G(\sigma)$, and $D_w \cap U_\sigma = \emptyset$ otherwise. By [CLS11, Theorem 3.2.6(c)(d)], for any face σ' of σ, $V(\sigma') \cap U_\sigma$ is the common zero locus of the coordinates corresponding to $G(\sigma')$, and $V(\sigma') \cap U_\sigma = \emptyset$ for any $\sigma' \in \Sigma$ that is not a face of σ. Since $G(\sigma)$ is part of a basis of N, the cone $\tau_{ij} + \mathrm{Cone}(v)$ is a face of σ if and only if $v \in G(\sigma)$. Thus, if $v \in G(\sigma)$, then D_v intersects $S_{ij} \cap U_\sigma$ transversally along $V(\tau_{ij} + \mathrm{Cone}(v)) \cap U_\sigma$. If $v \notin G(\sigma)$, then $D_v \cap S_{ij} \cap U_\sigma = \emptyset$, and $V(\tau_{ij} + \mathrm{Cone}(v)) \cap U_\sigma = \emptyset$

when $\tau_{ij} + \mathrm{Cone}(v) \in \Sigma$. Since the open sets U_σ, for $\sigma \in \Sigma$ having τ_{ij} as a face, cover S_{ij} [CLS11, Theorem 3.2.6(c)(d)], (3.1) holds. Since $\tau_{ij} + \mathrm{Cone}(x_j) = \mathrm{Cone}(P\backslash\{x_i\})$ and $\tau_{ij} + \mathrm{Cone}(x_i) = \mathrm{Cone}(P\backslash\{x_j\})$, (3.1) implies (2).

(3) Since no cone of Σ contains P, no cone of Σ contains both $\mathrm{Cone}(P\backslash\{x_i\})$ and x_i. Thus $D_{x_i} \cap C_i = \emptyset$ by [CLS11, Theorem 3.2.6(d)], and $(D_{x_i} \cdot C_i) = 0$. Similarly, $(D_{x_j} \cdot C_j) = 0$.

(4) Any cone of $\mathrm{Star}(\tau_{ij})$ is contained in a closed half-plane of $N_{ij} \otimes_{\mathbb{Z}} \mathbb{R}$ bounded by $\mathbb{R}\bar{x}_i$. Otherwise, some 2-dimensional cone $\bar{\sigma} \in \mathrm{Star}(\tau_{ij})$ intersects both open half-planes bounded by $\mathbb{R}\bar{x}_i$, and the interior of $\bar{\sigma}$ contains a non-zero point of $\mathbb{R}\bar{x}_i$. The rays contained in $\mathbb{R}\bar{x}_i$ are $\mathrm{Cone}(\bar{x}_i)$ and $\mathrm{Cone}(\bar{x}_j)$, and they belong to $\mathrm{Star}(\tau_{ij})$ since $\tau_{ij} + \mathrm{Cone}(x_i)$ and $\tau_{ij} + \mathrm{Cone}(x_j)$ belong to Σ. Thus $\mathrm{Cone}(\bar{x}_i)$ or $\mathrm{Cone}(\bar{x}_j)$ intersects $\bar{\sigma}$ in a ray that is not a face of $\bar{\sigma}$, which contradicts [CLS11, Definition 3.1.2(c)]. Since the kernel of $q_{ij} \otimes \mathbb{R}$ is $\mathbb{R}\bar{x}_i$, the map $q_{ij} \otimes \mathbb{R}$ sends any cone of $\mathrm{Star}(\tau_{ij})$ into $\mathbb{R}_{\geq 0}$ or $\mathbb{R}_{\leq 0}$, which are cones of the fan of $\mathbb{P}^1$ [CLS11, Example 3.1.11]. By [CLS11, Theorem 3.3.4(a)], q_{ij} induces a toric morphism $\pi_{ij} : S_{ij} \to \mathbb{P}^1$ whose restriction to the torus $T_{N_{ij}}$ is $q_{ij} \otimes \mathbb{C}^* : T_{N_{ij}} \to \mathbb{C}^*$. Since q_{ij} is surjective, $N_{ij} \cong \ker q_{ij} \oplus \mathbb{Z}$, so $q_{ij} \otimes \mathbb{C}^*$ is surjective. Since S_{ij} is projective, $\pi_{ij}(S_{ij})$ is a closed subset of $\mathbb{P}^1$ containing $\mathbb{C}^*$. Thus π_{ij} is surjective.

(5) Let t be the coordinate of the torus $\mathbb{C}^* \subset \mathbb{P}^1$, that is, the character corresponding to $1 \in \mathrm{Hom}(\mathbb{Z}, \mathbb{Z})$, and let $0, \infty \in \mathbb{P}^1$ be the torus-invariant points corresponding to $\mathbb{R}_{\geq 0}$ and $\mathbb{R}_{\leq 0}$, respectively. We regard 0 and ∞ as prime divisors on $\mathbb{P}^1$. Since the restriction of π_{ij} to $T_{N_{ij}}$ is $q_{ij} \otimes \mathbb{C}^*$, we have $\pi_{ij}^* t = \chi^{q_{ij}}$, where q_{ij} is regarded as an element of $\mathrm{Hom}(N_{ij}, \mathbb{Z})$. The elements of $G(\mathrm{Star}(\tau_{ij}))$ are the $\bar{w}$ with $w \in G(\Sigma)\backslash G(\tau_{ij})$ and $\tau_{ij} + \mathrm{Cone}(w) \in \Sigma$. Distinct such w have distinct images in N_{ij}: if $\bar{w} = \bar{w}'$ for two such elements w, w', then $w - w' \in \mathrm{Span}(\tau_{ij})$, so for $c > 0$ large enough, $w + c\sum_{y \in G(\tau_{ij})} y$ is contained in the relative interiors of both $\tau_{ij} + \mathrm{Cone}(w)$ and $\tau_{ij} + \mathrm{Cone}(w')$. By [CLS11, Definition 3.1.2(c)], $\tau_{ij} + \mathrm{Cone}(w) = \tau_{ij} + \mathrm{Cone}(w')$, and thus $w = w'$. For such w, let $E_{\bar{w}} := V(\tau_{ij} + \mathrm{Cone}(w))$, which is the torus-invariant prime divisor on S_{ij} corresponding to $\bar{w}$ [CLS11, Proposition 3.2.7]. We have $C_i = E_{\bar{x}_j}$ and $C_j = E_{\bar{x}_i}$. By [CLS11, Proposition 4.1.2], applied to $\mathbb{P}^1$ and to S_{ij}, we have

$$\pi_{ij}^*(0) - \pi_{ij}^*(\infty) = \pi_{ij}^* \operatorname{div}(t) = \operatorname{div}(\chi^{q_{ij}}) = \sum_{\bar{w} \in G(\mathrm{Star}(\tau_{ij}))} q_{ij}(\bar{w}) E_{\bar{w}}.$$

Since the divisors $\pi_{ij}^*(0) \geq 0$ and $\pi_{ij}^*(\infty) \geq 0$ have disjoint supports, we have

$$\pi_{ij}^*(0) = \sum_{q_{ij}(\bar{w}) > 0} q_{ij}(\bar{w}) E_{\bar{w}} \quad \text{and} \quad \pi_{ij}^*(\infty) = -\sum_{q_{ij}(\bar{w}) < 0} q_{ij}(\bar{w}) E_{\bar{w}}.$$

Since F_{ij} is the fiber of π_{ij} over a general point of $\mathbb{C}^*$, F_{ij} is linearly equivalent to $\pi_{ij}^*(0)$, and $F_{ij} \cap E_{\bar{w}} = \emptyset$ for any $\bar{w} \in G(\mathrm{Star}(\tau_{ij}))$ with $q_{ij}(\bar{w}) \neq 0$. Moreover, $(D_v \cdot F_{ij}) = (D_v|_{S_{ij}} \cdot F_{ij})_{S_{ij}}$ for any $v \in G(\Sigma)$.

Let $v \in G(\Sigma)\backslash P$. Then $v \notin G(\tau_{ij})$. If $\tau_{ij} + \mathrm{Cone}(v) \notin \Sigma$, then $D_v \cap S_{ij} = \emptyset$ by (3.1), so $D_v|_{S_{ij}} = 0$. Otherwise, $D_v|_{S_{ij}} = E_{\bar{v}}$ by (3.1). Since $v \neq x_i, x_j$, we have $\bar{v} \neq \bar{x}_i, \bar{x}_j$, so $\mathrm{Cone}(\bar{v})$ is a ray of $\mathrm{Star}(\tau_{ij})$ different from $\mathrm{Cone}(\bar{x}_i)$ and $\mathrm{Cone}(\bar{x}_j)$, which are the rays contained in $\mathbb{R}\bar{x}_i$. Thus $q_{ij}(\bar{v}) \neq 0$, and $F_{ij} \cap E_{\bar{v}} = \emptyset$. In both cases, $(D_v \cdot F_{ij}) = 0$.

By [CLS11, Theorem 10.4.4(b)], for any $\bar{w} \in G(\mathrm{Star}(\tau_{ij}))\backslash\{\bar{x}_j\}$, we have $(C_i \cdot E_{\bar{w}})_{S_{ij}} = 1$ if $\mathrm{Cone}(\bar{x}_j, \bar{w}) \in \mathrm{Star}(\tau_{ij})$, and $(C_i \cdot E_{\bar{w}})_{S_{ij}} = 0$ otherwise. Since $\mathrm{Star}(\tau_{ij})$ is complete [CLS11, Theorem 3.1.19(c)] and any cone of $\mathrm{Star}(\tau_{ij})$ is contained in a closed half-plane bounded by $\mathbb{R}\bar{x}_i$, there exists exactly one $\bar{w}_0 \in G(\mathrm{Star}(\tau_{ij}))$ such that $\mathrm{Cone}(\bar{x}_j, \bar{w}_0) \in \mathrm{Star}(\tau_{ij})$ and $q_{ij}(\bar{w}_0) > 0$. Since S_{ij} is smooth, $\{\bar{x}_j, \bar{w}_0\}$ is a basis of N_{ij} [CLS11, Theorem 3.1.19(a)]. As $q_{ij}(\bar{x}_j) = 0$ and q_{ij} is surjective, $q_{ij}(\bar{w}_0) = 1$. Since $C_i = E_{\bar{x}_j}$ and $q_{ij}(\bar{x}_j) = 0$, we obtain

$$(C_i \cdot F_{ij})_{S_{ij}} = (C_i \cdot \pi_{ij}^*(0))_{S_{ij}} = \sum_{q_{ij}(\bar{w}) > 0} q_{ij}(\bar{w})(C_i \cdot E_{\bar{w}})_{S_{ij}} = q_{ij}(\bar{w}_0) = 1.$$

Similarly, $(C_j \cdot F_{ij})_{S_{ij}} = 1$. By (2), $(D_{x_j} \cdot F_{ij}) = (C_i \cdot F_{ij})_{S_{ij}} = 1$ and $(D_{x_i} \cdot F_{ij}) = (C_j \cdot F_{ij})_{S_{ij}} = 1$.

It remains to consider $x_k \in G(\tau_{ij})$, that is, $k \neq i, j$. By [CLS11, Proposition 4.1.2], for any $u \in M$, we have

$$\operatorname{div}(\chi^u) = \sum_{v \in G(\Sigma)} \langle u, v \rangle D_v,$$

so $\sum_{v \in G(\Sigma)} \langle u, v \rangle (D_v \cdot F_{ij}) = 0$. Thus

$$\sum_{v \in G(\Sigma)} (D_v \cdot F_{ij}) v = 0.$$

Subtracting $\sum_{l=0}^{m} x_l = 0$ from the last equality and using $(D_v \cdot F_{ij}) = 0$ for $v \notin P$ and $(D_{x_i} \cdot F_{ij}) = (D_{x_j} \cdot F_{ij}) = 1$, we obtain

$$\sum_{0 \leq l \leq m, l \neq i, j} \big((D_{x_l} \cdot F_{ij}) - 1\big) x_l = 0.$$

Since $G(\tau_{ij}) = P \backslash \{x_i, x_j\}$ is linearly independent, $(D_{x_k} \cdot F_{ij}) = 1$ for any $k \neq i, j$. □

On a single surface S_{ij}, the sum $\sum_{k=0}^{m} (D_{x_k}^2 \cdot S_{ij})$ is the sum of terms ≤ 0 and a linear combination of the intersection numbers $(D_{x_k} \cdot C_i)$ and $(D_{x_k} \cdot C_j)$ on X, whose sign is not determined in general. After summing over all $0 \leq i < j \leq m$, the linear combinations add up to

$$\frac{3m-1}{4} \sum_{0 \leq i < j \leq m} ((C_i - C_j)^2)_{S_{ij}} \leq 0.$$

Proposition 3.3 concerns only the prime divisors D_{x_k}, since in Section 5 it is used for the variety of Construction-Proposition 4.4, where the other terms of the second Chern character are squares of pullbacks of prime divisors, and Lemma 2.5 bounds them separately.

Proposition 3.3. *Notation and conditions as in Set-up 3.1. Then*

$$\sum_{0 \leq i < j \leq m} \sum_{k=0}^{m} (D_{x_k}^2 \cdot S_{ij}) = \frac{3m-1}{4} \sum_{0 \leq i < j \leq m} \big((C_i - C_j)^2\big)_{S_{ij}} + \sum_{0 \leq i < j \leq m} \sum_{k=0}^{m} \Big(\big(D_{x_k}|_{S_{ij}} - \tfrac{1}{2}(C_i + C_j)\big)^2\Big)_{S_{ij}},$$

where $\big((C_i - C_j)^2\big)_{S_{ij}} \leq 0$ *and*

$$\big((D_{x_k}|_{S_{ij}} - \tfrac{1}{2}(C_i + C_j))^2\big)_{S_{ij}} \leq 0$$

for any $0 \leq i < j \leq m$ *and* $0 \leq k \leq m$. *In particular,*

$$\sum_{0 \leq i < j \leq m} \sum_{k=0}^{m} (D_{x_k}^2 \cdot S_{ij}) \leq 0.$$

Proof. By Construction-Proposition 3.2, for any $0 \leq i < j \leq m$, there exists a surjective homomorphism $q_{ij} : N_{ij} \to \mathbb{Z}$ whose kernel contains the image of x_i. For any $0 \leq i < j \leq m$, we fix such a q_{ij}, the induced morphism $\pi_{ij} : S_{ij} \to \mathbb{P}^1$, and a general fiber F_{ij} of π_{ij}. For any $0 \leq i \leq m$, let $c_i := \sum_{k=0}^{m} (D_{x_k} \cdot C_i)$. Since $m \geq 1$, Construction-Proposition 3.2(3) implies that $(D_{x_i} \cdot C_i) = 0$ for any $0 \leq i \leq m$.

Fix $0 \leq i < j \leq m$. In $\operatorname{Pic}(S_{ij}) \otimes_{\mathbb{Z}} \mathbb{Q}$, let $L_k := D_{x_k}|_{S_{ij}}$ for any $0 \leq k \leq m$, and let $C_{ij} := \frac{1}{2}(C_i + C_j)$. Let $s_{ij} := ((C_i - C_j)^2)_{S_{ij}}$. By Construction-Proposition 3.2(2)(5),

$$(C_i \cdot F_{ij})_{S_{ij}} = (D_{x_j} \cdot F_{ij}) = 1 \quad \text{and} \quad (C_j \cdot F_{ij})_{S_{ij}} = (D_{x_i} \cdot F_{ij}) = 1,$$

so $(C_{ij} \cdot F_{ij})_{S_{ij}} = 1$ and $((C_i - C_j) \cdot F_{ij})_{S_{ij}} = 1 - 1 = 0$. Since $C_i \cap C_j = \emptyset$ (Construction-Proposition 3.2(1)), we have $(C_i \cdot C_j)_{S_{ij}} = 0$. Thus

$$(C_{ij}^2)_{S_{ij}} = \frac{1}{4}\big((C_i^2)_{S_{ij}} + (C_j^2)_{S_{ij}}\big) = \frac{1}{4} s_{ij}.$$

For any $0 \le k \le m$, expanding $((L_k - C_{ij})^2)_{S_{ij}}$, we have

$$(L_k^2)_{S_{ij}} = 2(L_k \cdot C_{ij})_{S_{ij}} - (C_{ij}^2)_{S_{ij}} + ((L_k - C_{ij})^2)_{S_{ij}}.$$

Since $\mathcal{O}_X(D_{x_k})|_{C_i}$ is the restriction of $\mathcal{O}_X(D_{x_k})|_{S_{ij}}$ to C_i, we have $(L_k \cdot C_i)_{S_{ij}} = (D_{x_k} \cdot C_i)$. Similarly, $(L_k \cdot C_j)_{S_{ij}} = (D_{x_k} \cdot C_j)$. Moreover, $(L_k^2)_{S_{ij}} = (D_{x_k}^2 \cdot S_{ij})$. Summing over $0 \le k \le m$, we obtain

$$\sum_{k=0}^{m} (D_{x_k}^2 \cdot S_{ij}) = c_i + c_j - \frac{m+1}{4} s_{ij} + \sum_{k=0}^{m} ((L_k - C_{ij})^2)_{S_{ij}}. \tag{3.2}$$

By Construction-Proposition 3.2(2), $C_i = L_j$ and $C_j = L_i$. Thus

$$s_{ij} = (C_i^2)_{S_{ij}} + (C_j^2)_{S_{ij}} = (L_j \cdot C_i)_{S_{ij}} + (L_i \cdot C_j)_{S_{ij}} = (D_{x_j} \cdot C_i) + (D_{x_i} \cdot C_j). \tag{3.3}$$

For any $0 \le k \le m$, we have

$$((L_k - C_{ij}) \cdot F_{ij})_{S_{ij}} = (D_{x_k} \cdot F_{ij}) - 1 = 0$$

by Construction-Proposition 3.2(5). Since S_{ij} is a smooth projective surface and π_{ij} is surjective (Construction-Proposition 3.2(1)(4)), Lemma 2.5 implies that $s_{ij} \le 0$ and $((L_k - C_{ij})^2)_{S_{ij}} \le 0$ for any $0 \le k \le m$.

We sum over all $0 \le i < j \le m$. By (3.3) and since $(D_{x_i} \cdot C_i) = 0$ for any $0 \le i \le m$,

$$\sum_{0 \le i < j \le m} s_{ij} = \sum_{0 \le i < j \le m} \big((D_{x_j} \cdot C_i) + (D_{x_i} \cdot C_j)\big) = \sum_{i=0}^{m} \sum_{0 \le k \le m, k \ne i} (D_{x_k} \cdot C_i) = \sum_{i=0}^{m} c_i.$$

Since any $0 \le l \le m$ appears in exactly m of the pairs (i, j) with $0 \le i < j \le m$, we have

$$\sum_{0 \le i < j \le m} (c_i + c_j) = m \sum_{l=0}^{m} c_l = m \sum_{0 \le i < j \le m} s_{ij}.$$

Summing (3.2) over all $0 \le i < j \le m$, we obtain

$$\begin{aligned} \sum_{0 \le i < j \le m} \sum_{k=0}^{m} (D_{x_k}^2 \cdot S_{ij}) = \Big(m - \frac{m+1}{4}\Big) \sum_{0 \le i < j \le m} s_{ij} \\ + \sum_{0 \le i < j \le m} \sum_{k=0}^{m} \Big(\big(D_{x_k}|_{S_{ij}} - \tfrac{1}{2}(C_i + C_j)\big)^2 \Big)_{S_{ij}}. \end{aligned}$$

Since $m - \frac{m+1}{4} = \frac{3m-1}{4}$ and $s_{ij} = ((C_i - C_j)^2)_{S_{ij}}$, the equality in the proposition holds. Since $3m - 1 > 0$, $s_{ij} \le 0$, and $((D_{x_k}|_{S_{ij}} - \frac{1}{2}(C_i + C_j))^2)_{S_{ij}} \le 0$, each term on the right-hand side is non-positive, which implies the last inequality in the proposition. □

4. Closures of subtori

In this section, we prove Construction-Proposition 4.4. Given a smooth projective toric variety X whose fan has a centered primitive collection with at most $\dim X$ elements, Construction-Proposition 4.4 provides a toric morphism to X from a smooth projective toric variety to which the results of Section 3 can be applied.

For a centered primitive collection $P = \{x_0, \dots, x_m\}$ of the fan Σ of a smooth projective toric variety X of dimension d and for $0 \le i < j \le m$, the subvariety $V(\mathrm{Cone}(P \backslash \{x_i, x_j\}))$ of X is of dimension $d - m + 1$, so it is not a surface when $m \le d - 2$. If Σ has a cone γ of dimension $d - m - 1$ with $G(\gamma) \cap P = \emptyset$ such that $\gamma + \mathrm{Cone}(P \backslash \{x\}) \in \Sigma$ for any $x \in P$, then the computations of Section 3 can be carried out directly on $V(\gamma)$. We do not assume that such a cone exists, and instead use the closure in X of the subtorus of T_N corresponding to a sublattice of N of rank $m + 1$ containing $N \cap \mathrm{Span}(P)$, at the cost of passing to the normalization of the closure.

Construction-Proposition 4.1. Let X be a smooth projective toric variety of dimension d with fan Σ in $N_{\mathbb{R}}$, let $P = \{x_0, \dots, x_m\}$ be a centered primitive collection of Σ, let $N' \subset N$ be a sublattice of rank $m+1$, and let $N'_{\mathbb{R}} := N' \otimes_{\mathbb{Z}} \mathbb{R}$, a linear subspace of $N_{\mathbb{R}}$. Assume that the following hold.

(i) $m \leq d-1$.
(ii) N/N' is torsion-free.
(iii) $N \cap \mathrm{Span}(P) \subset N'$.

Then $\Sigma'_0 := \{\sigma \cap N'_{\mathbb{R}} \mid \sigma \in \Sigma\}$ is a complete fan in $N'_{\mathbb{R}}$, and the toric morphism $g : X_{\Sigma'_0} \to X$ induced by the inclusion $N' \subset N$ satisfies the following, where D^0_w denotes the torus-invariant prime divisor of $X_{\Sigma'_0}$ corresponding to $w \in G(\Sigma'_0)$.

(1) $X_{\Sigma'_0}$ is projective.
(2) $P \subset G(\Sigma'_0)$, $\mathrm{Cone}(P\backslash\{x\}) \in \Sigma'_0$ for any $x \in P$, and no cone of Σ'_0 contains P.
(3) For any $x \in P$, $g^*D_x = D^0_x$.
(4) For any $v \in G(\Sigma)\backslash P$, g^*D_v is a non-negative integral linear combination of the prime divisors D^0_w with $w \in G(\Sigma'_0)\backslash P$.

Proof. Since N/N' is torsion-free, $T_{N'}$ is a subtorus of T_N. Let $V \subset X$ be the closure of $T_{N'}$. By [CLS11, Proposition 11.2.14], Σ'_0 is a fan in $N'_{\mathbb{R}}$, and $X_{\Sigma'_0}$ is the normalization of V. Let $\nu : X_{\Sigma'_0} \to V$ be the normalization morphism. Since X is projective, Σ is complete [CLS11, Theorem 3.1.19(c)]. Thus $N'_{\mathbb{R}}$ is the union of the cones $\sigma \cap N'_{\mathbb{R}}$ with $\sigma \in \Sigma$, and Σ'_0 is complete.

(1) Since V is closed in the projective variety X, it is projective. The morphism ν is finite [Har77, Ch. II, Exercise 3.8] and dominant, hence surjective. Thus the pullback under ν of an ample line bundle on V is ample [Har77, Ch. III, Exercise 5.7(d)], and $X_{\Sigma'_0}$ is projective.

(2) Here $G(\Sigma'_0)$ is the set of primitive generators of the rays of Σ'_0 with respect to the lattice N'. By (iii), $P \subset N \cap \mathrm{Span}(P) \subset N'$. By Lemma 2.4(2), $x_1, \dots, x_m$ form part of a basis of N, hence a basis of $N \cap \mathrm{Span}(P)$, and $x_0 = -\sum_{k=1}^m x_k$. For any $Q \subsetneq P$, the cone $\mathrm{Cone}(Q)$ belongs to Σ and is contained in $\mathrm{Span}(P) \subset N'_{\mathbb{R}}$, so $\mathrm{Cone}(Q) = \mathrm{Cone}(Q) \cap N'_{\mathbb{R}}$ belongs to Σ'_0. In particular, $\mathrm{Cone}(P\backslash\{x\}) \in \Sigma'_0$ for any $x \in P$. The cones $\mathrm{Cone}(Q)$ with $Q \subsetneq P$ form the fan of $\mathbb{P}^m$ in $\mathrm{Span}(P)$ with respect to the basis $x_1, \dots, x_m$ of $N \cap \mathrm{Span}(P)$ [CLS11, Example 3.1.10], so their union is $\mathrm{Span}(P)$. By [CLS11, Definition 3.1.2(c)], the intersection of two cones of a fan is a face of both. Since a face of a cone containing a point of the relative interior of the cone is the cone itself, two cones of a fan whose relative interiors intersect are equal. Thus, if $w \in G(\Sigma'_0) \cap \mathrm{Span}(P)$ and $Q \subsetneq P$ is the subset such that the relative interior of $\mathrm{Cone}(Q)$ contains w, then $\mathrm{Cone}(w) = \mathrm{Cone}(Q)$, so $Q = \{x\}$ for some $x \in P$. Since any $x \in P$ is primitive in N, hence in N', it follows that $G(\Sigma'_0) \cap \mathrm{Span}(P) = P$. Finally, if a cone $\sigma \cap N'_{\mathbb{R}} \in \Sigma'_0$ with $\sigma \in \Sigma$ contains P, then σ contains x_0 and $-x_0 = \sum_{k=1}^m x_k$, which contradicts the strong convexity of σ.

(3) and (4) Let $w \in G(\Sigma'_0)\backslash P$, and let $\sigma_w \in \Sigma$ be the unique cone whose relative interior contains w. We show that $G(\sigma_w) \cap P = \emptyset$. Let U be a neighborhood of w in $\mathrm{Span}(\sigma_w)$ contained in σ_w. Then $U \cap N'_{\mathbb{R}}$ is a neighborhood of w in $\mathrm{Span}(\sigma_w) \cap N'_{\mathbb{R}}$ contained in $\sigma_w \cap N'_{\mathbb{R}}$. Since

$$\mathrm{Span}(\sigma_w \cap N'_{\mathbb{R}}) \subset \mathrm{Span}(\sigma_w) \cap N'_{\mathbb{R}},$$

the relative interior of $\sigma_w \cap N'_{\mathbb{R}}$ contains w. As $\mathrm{Cone}(w)$ and $\sigma_w \cap N'_{\mathbb{R}}$ are cones of Σ'_0 whose relative interiors contain w, they are equal. If $x \in G(\sigma_w) \cap P$, then $x \in \sigma_w \cap N'_{\mathbb{R}} = \mathrm{Cone}(w)$ by (iii), so $x = w$, since both are primitive in N'. This contradicts $w \notin P$, so $G(\sigma_w) \cap P = \emptyset$.

Now let $v \in G(\Sigma)$, and let φ_v be the support function of the Cartier divisor D_v. By [CLS11, Theorem 4.2.12(a)(b)], φ_v is linear on any cone of Σ, $\varphi_v(v) = -1$, and $\varphi_v(u) = 0$ for any $u \in G(\Sigma)\backslash\{v\}$. By [CLS11, Proposition 6.2.7], the support function of g^*D_v is the restriction of φ_v to $N'_{\mathbb{R}}$, so by [CLS11, Theorem 4.2.12(a)(b)],

$$g^*D_v = -\sum_{w \in G(\Sigma'_0)} \varphi_v(w) D^0_w.$$

For $x \in P$, we have $\varphi_v(x) = -1$ if $v = x$, and $\varphi_v(x) = 0$ otherwise. For $w \in G(\Sigma'_0)\backslash P$, since σ_w is smooth and $w \in N$ is contained in the relative interior of σ_w, we can write $w = \sum_{u \in G(\sigma_w)} a_u u$ with positive integers a_u. Thus $-\varphi_v(w) = a_v$ if $v \in G(\sigma_w)$, and $-\varphi_v(w) = 0$ otherwise. If $v \in P$, then $v \notin G(\sigma_w)$ for any $w \in G(\Sigma'_0)\backslash P$, so $g^*D_v = D^0_v$. This proves (3). If $v \notin P$, then the coefficients of g^*D_v at the divisors D^0_x with $x \in P$ vanish, and the other coefficients are non-negative integers. This proves (4). □

Depending on N', the toric variety $X_{\Sigma'_0}$ in Construction-Proposition 4.1 can be singular (Example 4.2).

Example 4.2. Let $N = \mathbb{Z}^3$ with standard basis e_1, e_2, e_3, and let X be the blow-up of $(\mathbb{P}^1)^3$ along the disjoint torus-invariant curves $V(\mathrm{Cone}(e_2, -e_3))$ and $V(\mathrm{Cone}(-e_2, e_3))$, that is, the toric variety of the fan Σ obtained from the fan of $(\mathbb{P}^1)^3$ by the star subdivisions at $e_2 - e_3$ and $e_3 - e_2$. Then $P = \{e_2 - e_3, e_3 - e_2\}$ is a centered primitive collection of Σ, with $m = 1$ and $d = 3$. Let

$$N' = \{(a, b, c) \in \mathbb{Z}^3 \mid 2a = b + c\}.$$

Then N' contains $N \cap \mathrm{Span}(P) = \mathbb{Z}(e_2 - e_3)$, and $N/N' \cong \mathbb{Z}$, so N' satisfies conditions (i)–(iii) of Construction-Proposition 4.1. The cone $\mathrm{Cone}(e_1, e_2, e_3)$ belongs to Σ, and

$$\mathrm{Cone}(e_1, e_2, e_3) \cap N'_{\mathbb{R}} = \mathrm{Cone}((1, 2, 0), (1, 0, 2)).$$

The vectors $(0, 1, -1)$ and $(1, 1, 1)$ form a basis of N', and

$$(1, 2, 0) = (1, 1, 1) + (0, 1, -1), \quad (1, 0, 2) = (1, 1, 1) - (0, 1, -1),$$

so $(1, 2, 0)$ and $(1, 0, 2)$, which are primitive in N and hence in N', generate a sublattice of index 2 in N'. Thus $\mathrm{Cone}((1, 2, 0), (1, 0, 2)) \in \Sigma'_0$ is not smooth [CLS11, Definition 1.2.16(a)], and $X_{\Sigma'_0}$ is singular [CLS11, Theorem 3.1.19(a)]. On the other hand, the sublattice $N'' := \mathbb{Z}e_1 \oplus \mathbb{Z}(e_2 - e_3)$ of N also satisfies conditions (i)–(iii), and

$$N''_{\mathbb{R}} := N'' \otimes_{\mathbb{Z}} \mathbb{R} = \{(a, b, c) \in \mathbb{R}^3 \mid b + c = 0\}.$$

The linear form ℓ on $\mathbb{R}^3$ with $\ell(a, b, c) = b + c$ is non-negative or non-positive on each cone $\sigma \in \Sigma$, so $\sigma \cap N''_{\mathbb{R}}$ is a face of σ. Hence the fan

$$\Sigma''_0 := \{\sigma \cap N''_{\mathbb{R}} \mid \sigma \in \Sigma\}$$

consists of the cones of Σ generated by subsets of $\{\pm e_1, \pm(e_2 - e_3)\}$, and $X_{\Sigma''_0} \cong \mathbb{P}^1 \times \mathbb{P}^1$.

We resolve $X_{\Sigma'_0}$ by star subdivisions, for any N'. If a star subdivision is taken at a vector in the relative interior of a cone of dimension at least two containing some $x \in P$, then the pullback of D^0_x acquires a new component with positive coefficient and is no longer a prime divisor, and the statement of [CLS11, Theorem 11.1.9] does not control the vectors at which the star subdivisions are taken. Since D^0_x is Cartier for any $x \in P$ (Construction-Proposition 4.1(3)), a cone $\sigma \in \Sigma'_0$ is smooth if the face $\mathrm{Cone}(G(\sigma)\backslash P)$ of σ is smooth, so it suffices to take star subdivisions at vectors contained in cones of Σ'_0 with no ray generated by an element of P (Lemma 4.3).

Lemma 4.3. *Let X be a toric variety with fan Σ in $N_{\mathbb{R}}$, let $P \subset G(\Sigma)$ be a subset, and let Σ_P be the set of cones $\sigma \in \Sigma$ such that $G(\sigma) \cap P = \emptyset$. Assume that D_x is Cartier for any $x \in P$. Then there exists a smooth fan Σ' in $N_{\mathbb{R}}$ refining Σ, given by finitely many star subdivisions of Σ at primitive vectors of N contained in $\bigcup_{\sigma \in \Sigma_P} \sigma$, such that the induced toric morphism $h : X' := X_{\Sigma'} \to X$ is projective and the following hold.*

(1) Any cone of Σ generated by a subset of P belongs to Σ'.
(2) For any $x \in P$, $h^*D_x = D'_x$, where D'_x is the torus-invariant prime divisor of X' corresponding to x.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we decompose the cones of Σ. For any $x \in P$, let φ_x be the support function of the Cartier divisor D_x. Then φ_x is linear on any cone of Σ, $\varphi_x(x) = -1$, and $\varphi_x(u) = 0$ for any $u \in G(\Sigma)\backslash\{x\}$ [CLS11, Theorem 4.2.12(a)(b)]. Fix $\sigma \in \Sigma$, and let $I_\sigma := G(\sigma) \cap P$. By [CLS11, Theorem 4.2.8(c)], for any $x \in I_\sigma$, there exists $\ell_{\sigma,x} \in M$ such that $\varphi_x(u) = -\ell_{\sigma,x}(u)$ for any $u \in \sigma$. In the rest of this step, we write ℓ_x for $\ell_{\sigma,x}$, which depends on σ. Then $\ell_x(x) = 1$ and $\ell_x(u) = 0$ for any $u \in G(\sigma)\backslash\{x\}$, so

$$\ell_x\Big(\sum_{u\in G(\sigma)} c_u u\Big) = c_x$$

for any real numbers c_u. Hence ℓ_x is non-negative on σ, and

$$\sigma_0 := \mathrm{Cone}(G(\sigma)\backslash I_\sigma) = \{u \in \sigma \mid \ell_x(u) = 0 \text{ for any } x \in I_\sigma\}$$

is a face of σ with $G(\sigma_0) = G(\sigma)\backslash I_\sigma$. In particular, $\sigma_0 \in \Sigma_P$ [CLS11, Definition 3.1.2(b)], and any face τ of σ with $G(\tau) \cap P = \emptyset$ is contained in σ_0, hence is a face of σ_0. Moreover, $\sigma = \mathrm{Cone}(I_\sigma) + \sigma_0$, and

$$N = \bigoplus_{x\in I_\sigma} \mathbb{Z}x \oplus N_0, \quad \text{where } N_0 := \{u \in N \mid \ell_x(u) = 0 \text{ for any } x \in I_\sigma\},$$

since $\ell_x(x') = 0$ for any $x \neq x'$ in I_σ, and $u - \sum_{x\in I_\sigma} \ell_x(u)x \in N_0$ for any $u \in N$. Let $I' \subset I_\sigma$, and let $\gamma \subset \sigma_0$ be a smooth cone. Then $G(\gamma) \subset \sigma_0 \cap N \subset N_0$, and $G(\gamma)$ is part of a basis of N since γ is smooth, hence part of a basis of the direct summand N_0. Thus $I' \cup G(\gamma)$ is part of a basis of N, and $\mathrm{Cone}(I') + \gamma$ is smooth [CLS11, Definition 1.2.16(a)]. Finally, since $G(\tau) \subset G(\sigma)$ for any face τ of a cone $\sigma \in \Sigma$, the set Σ_P is a subfan of Σ [CLS11, Definition 3.1.2(b)(c)].

Step 2. In this step, we study one star subdivision. Let $v \in N$ be a primitive vector contained in $\bigcup_{\gamma\in\Sigma_P} \gamma$, let $\Sigma^*(v)$ be the star subdivision of Σ at v [CLS11, §11.1, p. 515], and let h_v be the induced toric morphism $X_{\Sigma^*(v)} \to X$. We show that (a) any cone of Σ generated by a subset of P belongs to $\Sigma^*(v)$, (b) the cones $\tau \in \Sigma^*(v)$ with $G(\tau) \cap P = \emptyset$ are exactly the cones of the star subdivision $\Sigma_P^*(v)$ of Σ_P at v, and (c) for any $x \in P$, $h_v^*D_x$ is the torus-invariant prime divisor of $X_{\Sigma^*(v)}$ corresponding to x, and it is Cartier.

Since $v \in \gamma$ for some $\gamma \in \Sigma_P$, and φ_x is linear on γ and vanishes on $G(\gamma)$, we have $\varphi_x(v) = 0$ for any $x \in P$. In particular, $v \notin P$.

(a) Let $Q \subset P$ with $\mathrm{Cone}(Q) \in \Sigma$. The function $-\sum_{x\in Q} \varphi_x$ is linear on $\mathrm{Cone}(Q)$ and equal to 1 at any element of Q, so it is positive on $\mathrm{Cone}(Q)\backslash\{0\}$. Since it vanishes at $v \neq 0$, we have $v \notin \mathrm{Cone}(Q)$, so $\mathrm{Cone}(Q) \in \Sigma^*(v)$.

(b) Let $\sigma \in \Sigma$ with $v \in \sigma$. Since $\ell_{\sigma,x}(v) = -\varphi_x(v) = 0$ for any $x \in I_\sigma$, we have $v \in \sigma_0$ by **Step 1**. The cones of $\Sigma^*(v)$ not containing v are the cones of Σ not containing v, and those τ with $G(\tau) \cap P = \emptyset$ are the cones of Σ_P not containing v. A cone of $\Sigma^*(v)$ containing v is of the form $\tau + \mathrm{Cone}(v)$, where τ is a face of a cone $\sigma \in \Sigma$ with $v \in \sigma$ and $v \notin \tau$. Since τ and $\tau + \mathrm{Cone}(v)$ belong to the fan $\Sigma^*(v)$, τ is a face of $\tau + \mathrm{Cone}(v)$ [CLS11, Definition 3.1.2(c)]. As $v \notin P$, it follows that $G(\tau + \mathrm{Cone}(v)) \cap P = \emptyset$ if and only if $G(\tau) \cap P = \emptyset$, that is, if and only if τ is a face of σ_0 (**Step 1**). Since $v \in \sigma_0 \in \Sigma_P$ for any such σ, and $\gamma_0 = \gamma$ for any $\gamma \in \Sigma_P$, the cones $\tau + \mathrm{Cone}(v) \in \Sigma^*(v)$ with $G(\tau + \mathrm{Cone}(v)) \cap P = \emptyset$ are exactly the cones of $\Sigma_P^*(v)$ containing v. This proves (b).

(c) Let $x \in P$. By (a) with $Q = \{x\}$, $\mathrm{Cone}(x) \in \Sigma^*(v)$, so $x \in G(\Sigma^*(v))$. By [CLS11, Proposition 6.2.7], φ_x is the support function of $h_v^*D_x$, and so by [CLS11, Theorem 4.2.12(a)(b)], the coefficient of $h_v^*D_x$ at the torus-invariant prime divisor corresponding to $w \in G(\Sigma^*(v))$ is $-\varphi_x(w)$. Since $G(\Sigma^*(v)) \subset G(\Sigma) \cup \{v\}$ and $\varphi_x(v) = 0$, we have $-\varphi_x(x) = 1$ and $-\varphi_x(w) = 0$ for any $w \in G(\Sigma^*(v))\backslash\{x\}$. Thus $h_v^*D_x$ is the torus-invariant prime divisor of $X_{\Sigma^*(v)}$ corresponding to x, and it is Cartier since it is the pullback of the Cartier divisor D_x.

Step 3. In this step, we construct Σ' and conclude the proof. By [CLS11, Theorem 11.1.9(a)(c)] applied to the fan Σ_P, there exist primitive vectors $v_1, \dots, v_k \in N$ such that, if $\Sigma_P^{(0)} := \Sigma_P$ and $\Sigma_P^{(i)}$ is the star subdivision of $\Sigma_P^{(i-1)}$ at $v_i \in \bigcup_{\gamma \in \Sigma_P^{(i-1)}} \gamma$ for $1 \le i \le k$, then $\Sigma_P^{(k)}$ is smooth. Since star subdivisions do not change the support of a fan, $v_i \in \bigcup_{\gamma \in \Sigma_P} \gamma$ for any i. Let $\Sigma^{(0)} := \Sigma$, and for $1 \le i \le k$, let $\Sigma^{(i)}$ be the star subdivision of $\Sigma^{(i-1)}$ at v_i, and let $h_i : X_{\Sigma^{(i)}} \to X_{\Sigma^{(i-1)}}$ be the induced toric morphism.

We show by induction on $0 \le i \le k$ that the following hold, which we call (a)–(c) for i: (a) any cone of Σ generated by a subset of P belongs to $\Sigma^{(i)}$; (b) the cones $\tau \in \Sigma^{(i)}$ with $G(\tau) \cap P = \emptyset$ are exactly the cones of $\Sigma_P^{(i)}$; (c) for any $x \in P$, $(h_1 \circ \cdots \circ h_i)^* D_x$ is the torus-invariant prime divisor of $X_{\Sigma^{(i)}}$ corresponding to x, and it is Cartier. For $i = 0$, there is nothing to prove. For $i \ge 1$, by (a)–(c) for $i-1$, $X_{\Sigma^{(i-1)}}$ and P satisfy the assumptions of the lemma, and v_i is contained in a cone $\tau \in \Sigma^{(i-1)}$ with $G(\tau) \cap P = \emptyset$. Thus **Steps 1 and 2** with $X_{\Sigma^{(i-1)}}$ and v_i in place of X and v, together with (a)–(c) for $i-1$, imply (a)–(c) for i.

Let $\Sigma' := \Sigma^{(k)}$, $X' := X_{\Sigma'}$, and $h := h_1 \circ \cdots \circ h_k$. Then Σ' refines Σ and is given by finitely many star subdivisions at primitive vectors of N contained in $\bigcup_{\sigma \in \Sigma_P} \sigma$. Each h_i is projective [CLS11, Proposition 11.1.6(c)], so h is projective [CLS11, Proposition 7.0.5(a)]. (a) and (c) for $i = k$ imply (1) and (2). It remains to show that Σ' is smooth. Let $\sigma' \in \Sigma'$. Since D'_x is Cartier for any $x \in P$, **Step 1** with X' in place of X shows that there exists a face σ'_0 of σ' with $G(\sigma'_0) \cap P = \emptyset$ and $\sigma' = \mathrm{Cone}(G(\sigma') \cap P) + \sigma'_0$, and that σ' is smooth if σ'_0 is smooth. By (b) for $i = k$, σ'_0 is a cone of the smooth fan $\Sigma_P^{(k)}$. Thus σ' is smooth. □

Construction-Proposition 4.4. Let X be a smooth projective toric variety of dimension d with fan Σ in $N_{\mathbb{R}}$, let $P = \{x_0, \dots, x_m\}$ be a centered primitive collection of Σ, let $N' \subset N$ be a sublattice of rank $m+1$, and let $N'_{\mathbb{R}} := N' \otimes_{\mathbb{Z}} \mathbb{R}$, a linear subspace of $N_{\mathbb{R}}$. Assume that the following hold.

(i) $m \le d-1$.
(ii) N/N' is torsion-free.
(iii) $N \cap \mathrm{Span}(P) \subset N'$.

Then there exists a smooth fan Σ' in $N'_{\mathbb{R}}$ refining $\{\sigma \cap N'_{\mathbb{R}} \mid \sigma \in \Sigma\}$ such that $X' := X_{\Sigma'}$ is projective and the toric morphism $f : X' \to X$ induced by the inclusion $N' \subset N$ satisfies the following, where D'_w denotes the torus-invariant prime divisor of X' corresponding to $w \in G(\Sigma')$.

(1) P is a centered primitive collection of Σ'.
(2) For any $x \in P$, $f^* D_x = D'_x$.
(3) For any $v \in G(\Sigma) \backslash P$, $f^* D_v = \sum_{w \in G(\Sigma') \backslash P} a_{v,w} D'_w$ for some non-negative integers $a_{v,w}$.
(4) For any $0 \le i < j \le m$, f restricts to a birational morphism from $V(\mathrm{Cone}(P \backslash \{x_i, x_j\})) \subset X'$ onto a surface in X.

Proof. We use the notation Σ'_0, g, and D^0_w of Construction-Proposition 4.1. By Construction-Proposition 4.1(2)(3), $P \subset G(\Sigma'_0)$, and for any $x \in P$, the divisor $D^0_x = g^* D_x$ is Cartier, since it is the pullback of a Cartier divisor. Thus, by Lemma 4.3 applied to $X_{\Sigma'_0}$, Σ'_0, and P, with N' and D^0_x in place of N and D_x, there exists a smooth fan Σ' in $N'_{\mathbb{R}}$ refining Σ'_0 such that the induced toric morphism $h : X' = X_{\Sigma'} \to X_{\Sigma'_0}$ is projective and (1) and (2) of Lemma 4.3 hold. Let $f := g \circ h$. Then f is the toric morphism induced by the inclusion $N' \subset N$, and X' is projective by Construction-Proposition 4.1(1) and [CLS11, Proposition 7.0.5(a)(c)].

(1) By Construction-Proposition 4.1(2), $\mathrm{Cone}(x)$ and $\mathrm{Cone}(P \backslash \{x\})$ belong to Σ'_0 for any $x \in P$, so they belong to Σ' by Lemma 4.3(1). In particular, $P \subset G(\Sigma')$. Since Σ' refines Σ'_0 and no cone of Σ'_0 contains P, no cone of Σ' contains P, so $\mathrm{Cone}(P) \notin \Sigma'$. As Σ' is smooth and $\sum_{x \in P} x = 0$, P is a centered primitive collection of Σ' (Definition 2.3 with N' in place of N).

(2) For any $x \in P$, Construction-Proposition 4.1(3) and Lemma 4.3(2) imply

$$f^* D_x = h^* g^* D_x = h^* D^0_x = D'_x.$$

(3) Fix $v \in G(\Sigma)\backslash P$, and let φ be the support function of the Cartier divisor g^*D_v. By Construction-Proposition 4.1(4) and [CLS11, Theorem 4.2.12(b)], $\varphi(w) \leq 0$ for any $w \in G(\Sigma'_0)$, and $\varphi(x) = 0$ for any $x \in P$. Since φ is linear on any cone of Σ'_0, and any cone of Σ' is contained in a cone of Σ'_0, we have $\varphi(w) \leq 0$ for any $w \in G(\Sigma')$. By [CLS11, Proposition 6.2.7], φ is also the support function of $h^*g^*D_v = f^*D_v$, so by [CLS11, Theorem 4.2.12(a)(b)],

$$f^*D_v = -\sum_{w\in G(\Sigma')} \varphi(w)D'_w = \sum_{w\in G(\Sigma')\backslash P} a_{v,w}D'_w, \quad \text{where } a_{v,w} := -\varphi(w).$$

The numbers $a_{v,w}$ are non-negative, and they are integers since f^*D_v is a divisor.

(4) Fix $0 \leq i < j \leq m$, and recall that $\tau_{ij} = \mathrm{Cone}(P\backslash\{x_i, x_j\})$ (Set-up 3.1). For $\tau \in \Sigma'$, we write $V_{X'}(\tau)^\circ$ for the orbit of X' corresponding to τ and $V_{X'}(\tau)$ for the closure of $V_{X'}(\tau)^\circ$ in X'. For $\tau \in \Sigma$, we write $V_X(\tau)^\circ$ for the orbit of X corresponding to τ and $V_X(\tau)$ for the closure of $V_X(\tau)^\circ$ in X. By (1) and Definition 2.3, τ_{ij} is a face of the cone $\mathrm{Cone}(P\backslash\{x_i\})$, which belongs to both Σ' and Σ, so τ_{ij} is a cone of both Σ' and Σ. Since $\mathrm{Cone}(P\backslash\{x_i\})$ is smooth, $\dim \tau_{ij} = m-1$, so $V_{X'}(\tau_{ij})$ is a surface [CLS11, Theorem 3.2.6(b)]. By (iii), $N \cap \mathrm{Span}(\tau_{ij}) \subset N \cap \mathrm{Span}(P) \subset N'$, so $N' \cap \mathrm{Span}(\tau_{ij}) = N \cap \mathrm{Span}(\tau_{ij})$. Thus the homomorphism

$$\psi : N'/(N' \cap \mathrm{Span}(\tau_{ij})) \to N/(N \cap \mathrm{Span}(\tau_{ij}))$$

induced by the inclusion $N' \subset N$ is injective, and the cokernel of ψ is isomorphic to N/N', which is torsion-free by (ii). Hence the image of ψ is a direct summand, and the homomorphism of tori induced by ψ is a closed embedding.

By [CLS11, Lemma 3.2.5], the orbits $V_{X'}(\tau_{ij})^\circ$ and $V_X(\tau_{ij})^\circ$ are the tori of the lattices $N'/(N' \cap \mathrm{Span}(\tau_{ij}))$ and $N/(N \cap \mathrm{Span}(\tau_{ij}))$, respectively. Here we use that, since τ_{ij} is smooth, $\tau_{ij} \cap N'$ and $\tau_{ij} \cap N$ generate the sublattices $N' \cap \mathrm{Span}(\tau_{ij})$ and $N \cap \mathrm{Span}(\tau_{ij})$ of N' and N, respectively. The morphism f maps the distinguished point of $V_{X'}(\tau_{ij})^\circ$ to the distinguished point of $V_X(\tau_{ij})^\circ$ [CLS11, Lemma 3.3.21(a)], and f is equivariant with respect to $T_{N'} \to T_N$. Thus the restriction of f to $V_{X'}(\tau_{ij})^\circ$ is the homomorphism of tori induced by ψ, which is a closed embedding into $V_X(\tau_{ij})^\circ$. Let $\sigma' \in \Sigma'$ be a cone containing τ_{ij} as a proper face. The minimal cone of Σ containing σ' is not τ_{ij}, since it contains $\sigma' \supsetneq \tau_{ij}$, so f maps the orbit $V_{X'}(\sigma')^\circ$ into an orbit of X other than $V_X(\tau_{ij})^\circ$ [CLS11, Lemma 3.3.21(b)]. Since $V_{X'}(\tau_{ij})$ is the union of $V_{X'}(\tau_{ij})^\circ$ and the orbits $V_{X'}(\sigma')^\circ$ for such σ' [CLS11, Theorem 3.2.6(d)], the preimage of $V_X(\tau_{ij})^\circ$ under the restriction of f to $V_{X'}(\tau_{ij})$ is $V_{X'}(\tau_{ij})^\circ$. Let $S \subset X$ be the image of $V_{X'}(\tau_{ij})$ under f, which is closed since X' is projective. Since f is continuous and maps $V_{X'}(\tau_{ij})^\circ$ into $V_X(\tau_{ij})^\circ$, S is contained in the closure $V_X(\tau_{ij})$ of $V_X(\tau_{ij})^\circ$, in which $V_X(\tau_{ij})^\circ$ is open [CLS11, Theorem 3.2.6(c)(d)]. Then f restricts to an isomorphism from $V_{X'}(\tau_{ij})^\circ$ onto the intersection of S with $V_X(\tau_{ij})^\circ$, which is a non-empty open subset of S. Thus f restricts to a birational morphism from $V_{X'}(\tau_{ij})$ onto the surface S in X. □

5. Proof of the main theorem

In this section, we prove Proposition 5.1 by applying the results of Section 3 to the toric variety X' of Construction-Proposition 4.4, and then we prove Theorem 1.3 by combining Proposition 5.1 with Lemma 2.4.

Proposition 5.1. *Let X be a smooth projective toric variety of dimension d with fan Σ in $N_\mathbb{R}$, and let $P = \{x_0, \ldots, x_m\}$ be a centered primitive collection of Σ. Assume that $m \leq d-1$. Then there exists a surface $S \subset X$ such that $\mathrm{ch}_2(T_X) \cdot S \leq 0$.*

Proof. Since $d \geq m+1 \geq 1$, Lemma 2.4(2) implies that $m \geq 1$ and that $x_1, \ldots, x_m$ form part of a basis $x_1, \ldots, x_m, e_{m+1}, \ldots, e_d$ of N. Let

$$N' := \bigoplus_{k=1}^{m} \mathbb{Z}x_k \oplus \mathbb{Z}e_{m+1} \subset N.$$

Then $\operatorname{rank} N' = m+1$, and N' satisfies conditions (i)–(iii) of Construction-Proposition 4.4. Condition (i) is our assumption, and (ii) holds since $N/N' \cong \mathbb{Z}^{d-m-1}$. For (iii), since P is centered, $x_0 = -\sum_{k=1}^m x_k$, so $\operatorname{Span}(P) = \operatorname{Span}(\{x_1, \dots, x_m\})$. Since $x_1, \dots, x_m, e_{m+1}, \dots, e_d$ form a basis of N, it follows that

$$N \cap \operatorname{Span}(P) = \bigoplus_{k=1}^m \mathbb{Z} x_k \subset N'.$$

By Construction-Proposition 4.4 applied to X, P, and N', we obtain a smooth fan Σ' in $N'_{\mathbb{R}}$, the projective toric variety $X' = X_{\Sigma'}$, and the toric morphism $f : X' \to X$. Since X' is of dimension $\operatorname{rank} N' = m+1$, by Construction-Proposition 4.4(1), X', Σ', and P satisfy the conditions of Set-up 3.1 with N' in place of N. For any $0 \le i < j \le m$, let $S'_{ij} := V(\tau_{ij}) \subset X'$, let $\pi'_{ij} : S'_{ij} \to \mathbb{P}^1$ be the toric morphism induced by one of the homomorphisms $q_{ij} : N'/(N' \cap \operatorname{Span}(\tau_{ij})) \to \mathbb{Z}$ of Construction-Proposition 3.2 applied to X', and let F'_{ij} be a general fiber of π'_{ij}. By Construction-Proposition 3.2(1)(4) applied to X', S'_{ij} is a smooth projective surface, and π'_{ij} is surjective. Let $W_{ij} := f(S'_{ij})$. By Construction-Proposition 4.4(4), W_{ij} is a surface in X, and $f|_{S'_{ij}} : S'_{ij} \to W_{ij}$ is birational.

By [CLS11, Proposition 13.1.2(a)] and [Har77, Appendix A, §4],

$$2\operatorname{ch}_2(T_X) = \sum_{v \in G(\Sigma)} D_v^2.$$

Let $0 \le i < j \le m$. Taking degrees on W_{ij} and using [KM98, Proposition 1.35(1)(6)] for the birational morphism $f|_{S'_{ij}} : S'_{ij} \to W_{ij}$, we obtain

$$2\operatorname{ch}_2(T_X) \cdot W_{ij} = \sum_{v \in G(\Sigma)} (D_v^2 \cdot W_{ij}) = \sum_{v \in G(\Sigma)} \big((f^* D_v)^2 \cdot S'_{ij}\big).$$

We split the last sum into the terms with $v \in P$ and the terms with $v \in G(\Sigma) \backslash P$. For any $x \in P$, $f^* D_x = D'_x$ by Construction-Proposition 4.4(2). Let $v \in G(\Sigma) \backslash P$. By Construction-Proposition 4.4(3), we can write $f^* D_v = \sum_{w \in G(\Sigma') \backslash P} a_{v,w} D'_w$ with integers $a_{v,w}$. By Construction-Proposition 3.2(5) applied to X', $(D'_w \cdot F'_{ij}) = 0$ for any $w \in G(\Sigma') \backslash P$. Thus, by [KM98, Proposition 1.35(1)],

$$\big(f^* D_v|_{S'_{ij}} \cdot F'_{ij}\big)_{S'_{ij}} = (f^* D_v \cdot F'_{ij}) = \sum_{w \in G(\Sigma') \backslash P} a_{v,w} (D'_w \cdot F'_{ij}) = 0.$$

By Lemma 2.5 applied to S'_{ij}, π'_{ij}, and a $\mathbb{Q}$-divisor on S'_{ij} with class $f^* D_v|_{S'_{ij}}$, and by [KM98, Proposition 1.35(1)] again, we have

$$\big((f^* D_v)^2 \cdot S'_{ij}\big) = \big((f^* D_v|_{S'_{ij}})^2\big)_{S'_{ij}} \le 0.$$

Therefore,

$$2\operatorname{ch}_2(T_X) \cdot W_{ij} \le \sum_{k=0}^m \big((D'_{x_k})^2 \cdot S'_{ij}\big).$$

Summing over all $0 \le i < j \le m$, by Proposition 3.3 applied to X', we obtain

$$2 \sum_{0 \le i < j \le m} \operatorname{ch}_2(T_X) \cdot W_{ij} \le \sum_{0 \le i < j \le m} \sum_{k=0}^m \big((D'_{x_k})^2 \cdot S'_{ij}\big) \le 0.$$

Thus $\operatorname{ch}_2(T_X) \cdot W_{ij} \le 0$ for some $0 \le i < j \le m$. □

Proof of Theorem 1.3. Let Σ be the fan of X in $N_{\mathbb{R}}$. By Lemma 2.4(1), Σ has a centered primitive collection $P = \{x_0, \dots, x_m\}$, and $1 \le m \le d$ by Lemma 2.4(2). Suppose that $m \le d-1$. Then Proposition 5.1 implies that $\operatorname{ch}_2(T_X) \cdot S \le 0$ for some surface $S \subset X$, a contradiction. Thus $m = d$, so P has $d+1$ elements, and $X \cong \mathbb{P}^d$ by Lemma 2.4(3). □

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